

GOOGLE GEMINI POWERED LEARNING ASSISTANT

PROJECT DOCUMENTATION

EduGenie – AI-Assisted Learning Platform

Role	Name
Team Lead	Santhiya. N
Team Member 1	Dharshini. R
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Team Member 3	Santhana Sharvani. S
Team Member 4	Swathi Lakshmi. S

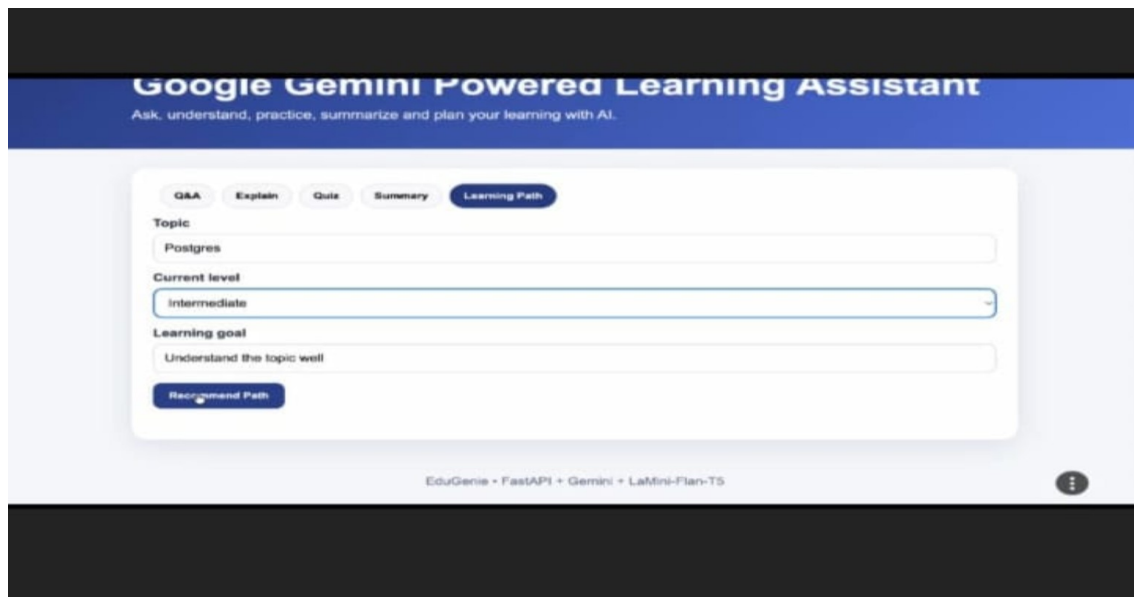


Figure 1: Google Gemini Powered Learning Assistant interface

1. INTRODUCTION

Google Gemini Powered Learning Assistant is an AI-assisted educational web application designed to support students in understanding, practising, summarizing and planning learning activities. The interface provides dedicated modules for Q&A, Explain, Quiz, Summary and Learning Path.

The demonstrated application uses a topic-driven workflow. The Learning Path screen shows the topic Postgres, an Intermediate level and the goal “Understand the topic well”, after which the system produces a structured learning path.

2. OBJECTIVES

- Provide an AI-powered learning assistant through a simple web interface.
- Allow users to ask questions and receive topic-based answers.
- Explain technical topics in an easy-to-understand format.
- Generate practice quizzes and display quiz results.
- Create concise summaries of learning topics.
- Generate a structured learning path based on topic, current level and learning goal.
- Integrate the web interface with an AI service for live response generation.

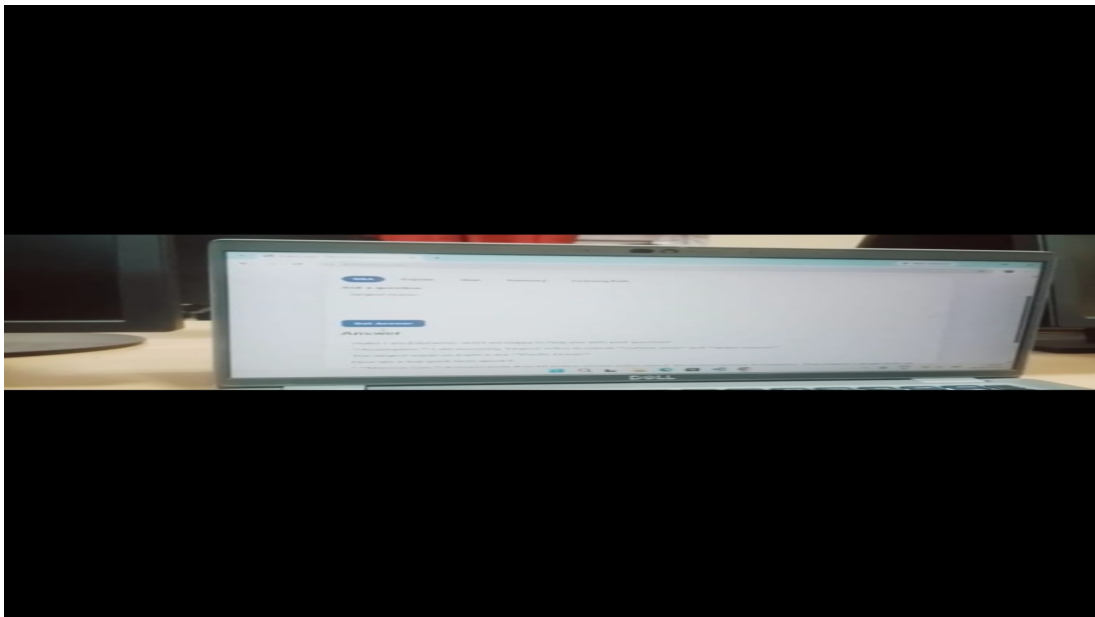


Figure 2: Q&A; module showing an AI-generated answer

3. EPIC – PREREQUISITES

The project requires a Python-based development environment and a web application layer capable of communicating with an AI model. The demonstrated application identifies FastAPI, Gemini and LangChain in its technology stack.

- Python development environment
- FastAPI for backend API services
- Google Gemini for AI-powered generation
- LangChain for AI/application integration support
- Browser-based frontend for user interaction
- Internet/API connectivity for live AI responses
- Supporting libraries for application execution

4. EPIC 1 – MODEL SELECTION AND ARCHITECTURE

4.1 Select AI Model

Google Gemini is selected as the AI model for the learning assistant. The application is presented as a Google Gemini Powered Learning Assistant and uses Gemini-based generation for learning content.

4.2 Architecture

The architecture follows a frontend–backend–AI service flow. The user interacts with the web interface, the backend handles application requests, and the AI service generates the requested learning content. The generated response is then displayed in the interface.

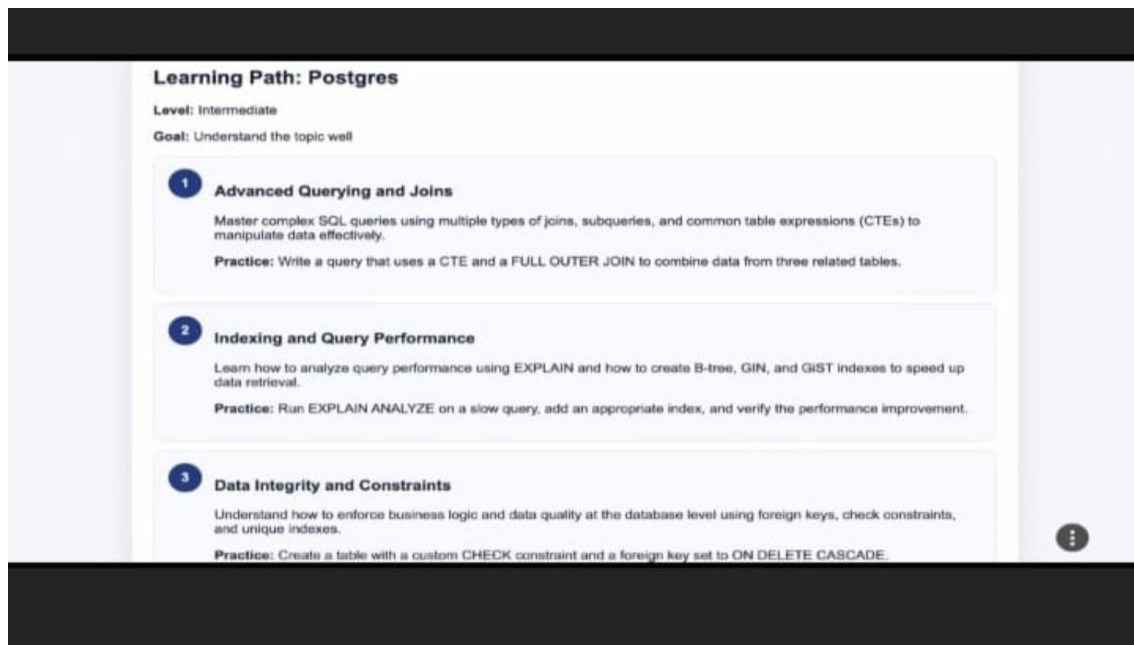


Figure 3: Learning Path input screen

5. EPIC 2 – CORE FUNCTIONALITIES AND DEVELOPMENT

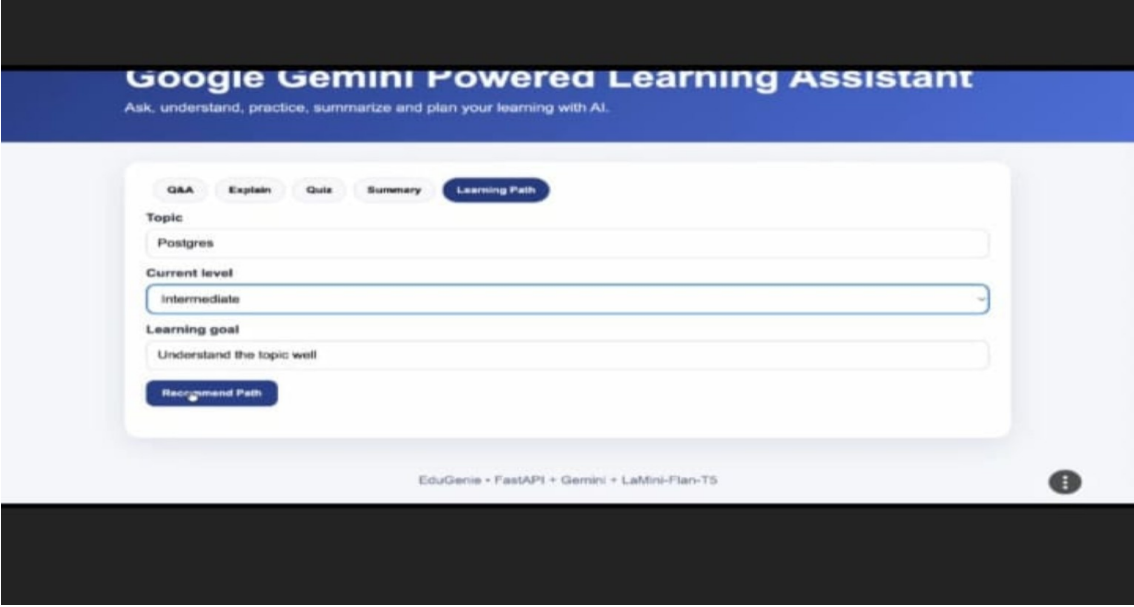
The core functionality is divided into learning modules. Each module accepts a user requirement and produces a corresponding AI-assisted learning output.

5.1 Module Implementation

- Q&A; Module – accepts a question and produces an answer.
- Explain Module – provides a topic explanation for the learner.
- Quiz Module – generates multiple-choice practice questions.
- Summary Module – presents a concise topic summary.
- Learning Path Module – creates a sequence of recommended learning areas.

5.2 Learning Path Module

For the demonstrated Postgres example, the generated learning path contains Advanced Querying and Joins, Indexing and Query Performance, and Data Integrity and Constraints.



The screenshot displays the 'Google Gemini Powered Learning Assistant' interface. At the top, a blue header contains the title and the tagline 'Ask, understand, practice, summarize and plan your learning with AI.' Below this, a navigation bar features five tabs: 'Q&A', 'Explain', 'Quiz', 'Summary', and 'Learning Path', with the latter being the active tab. The main form area includes four input fields: 'Topic' (containing 'Postgres'), 'Current level' (a dropdown menu set to 'Intermediate'), 'Learning goal' (containing 'Understand the topic well'), and a 'Recommend Path' button at the bottom. The footer of the interface shows the text 'EduGenie • FastAPI • Gemini • LaMini-Plan-T5' and a small circular icon with three dots.

Figure 4: Learning Path output for Postgres

6. EPIC 2 – BACKEND API AND FASTAPI

6.1 Backend API

FastAPI is used as the backend API layer identified in the demonstrated application. It provides the service layer between the web interface and the AI generation workflow.

6.2 Request and Response Flow

- User selects a learning module.
- The frontend collects the required topic, question or learning details.
- The request is passed to the backend API.
- The backend communicates with the Gemini-based generation workflow.
- The generated learning content is returned to the frontend.
- The result is displayed to the user.

6.3 AI Integration

The application integrates Google Gemini into the learning workflow so that generated content can be returned dynamically for different topics and learning tasks. The interface also identifies LangChain as part of the project technology stack.

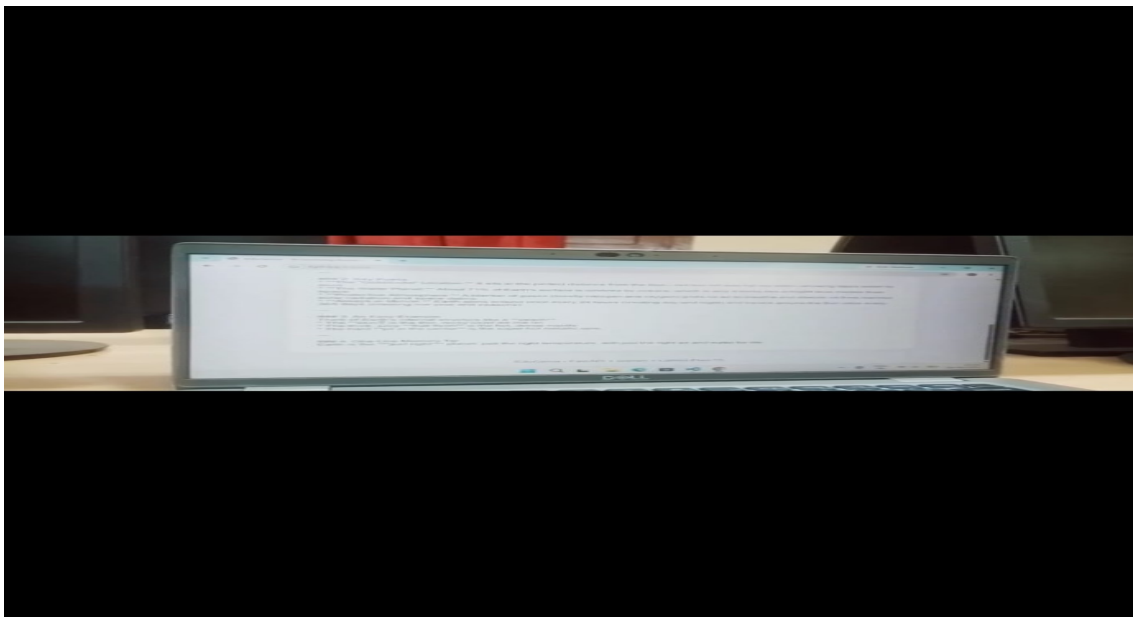


Figure 5: Generated explanatory learning content

7. EPIC 3 – FRONTEND DEVELOPMENT

7.1 Built Web Interface

The application provides a browser-based interface with clear navigation between Q&A, Explain, Quiz, Summary and Learning Path.

- Simple navigation between learning modules
- Topic/question input areas
- Action buttons for generating content
- Display area for generated results
- Quiz question and answer interface
- Learning Path input fields for topic, level and learning goal

7.2 Live Integration

The screenshots demonstrate the frontend receiving generated content and displaying it directly in the web interface, showing the application workflow operating through the browser.

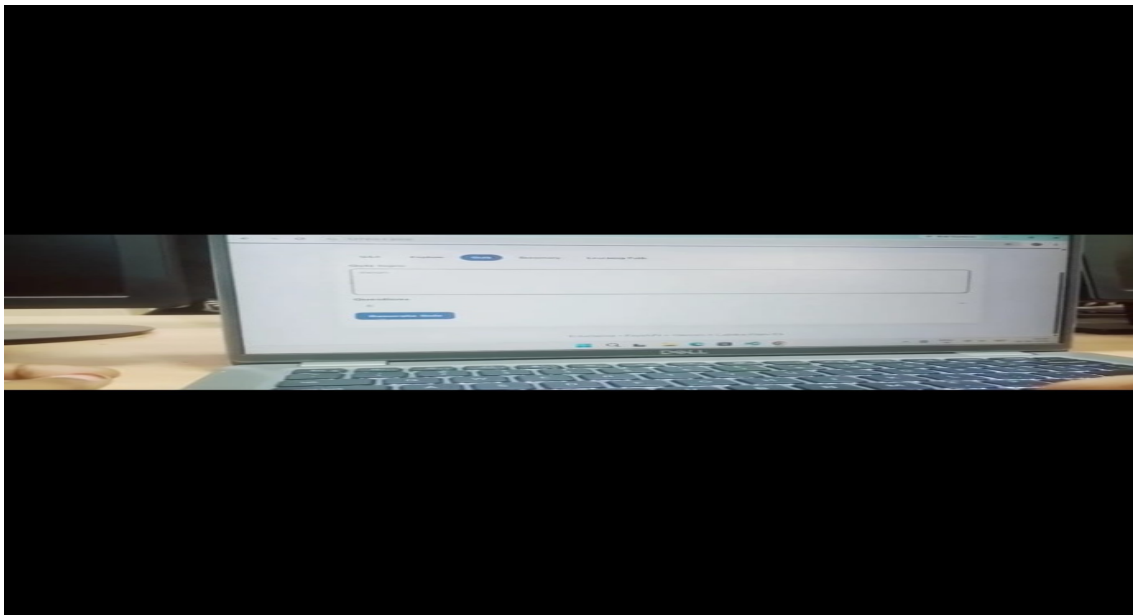


Figure 6: Quiz module used to generate a practice quiz

8. EPIC 3 – FRONTEND OUTPUTS

8.1 Quiz Output

The Quiz module presents generated multiple-choice questions to the learner. The demonstrated screen contains questions related to the Moon with selectable answer options.

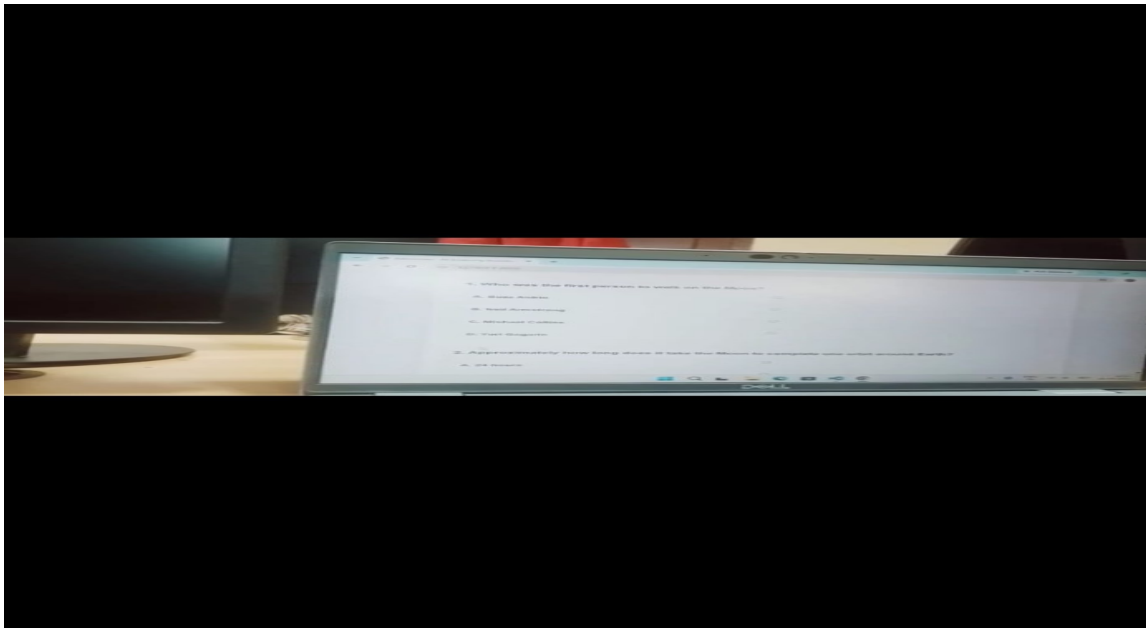


Figure 7: Generated quiz questions and answer options

8.2 Quiz Result

The demonstrated quiz result shows a score of 2 out of 5. The screen also indicates an answer selection and displays the correct answer for review, providing immediate feedback.

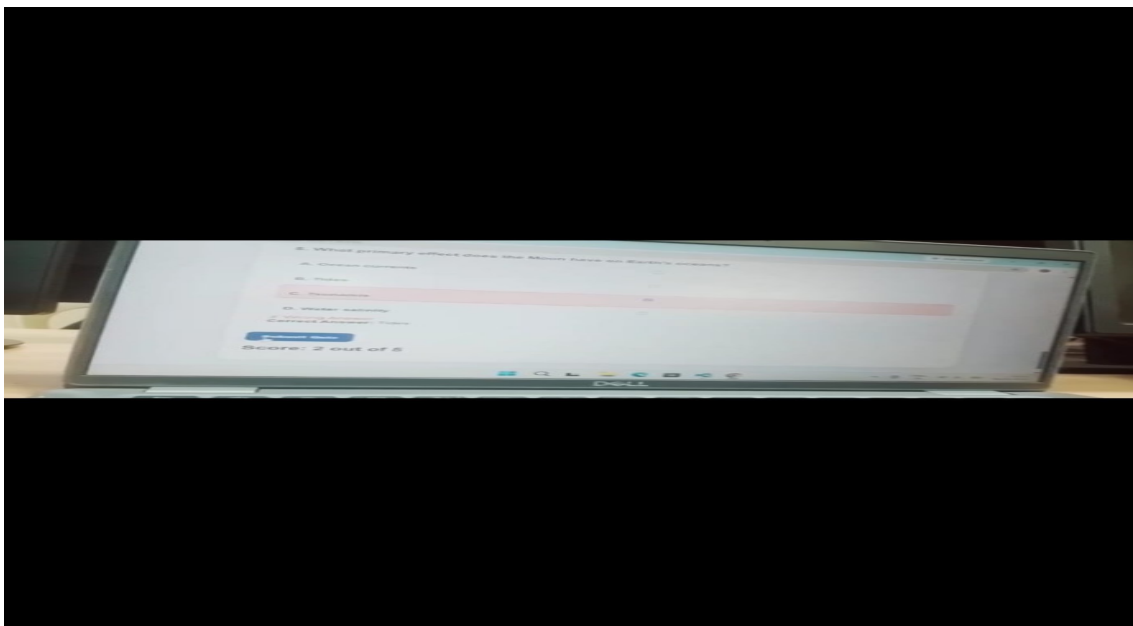


Figure 8: Quiz result and feedback screen

9. EPIC 4 – TESTING AND DEPLOYMENT

9.1 Run Locally

The application was demonstrated through a browser running on a laptop. The supplied evidence shows the learning workflow being executed through the web interface. No verified cloud deployment URL is provided in the supplied material, so this documentation records the demonstrated execution.

9.2 Functional Testing

- Q&A; module – generated an answer for a user request.
- Explain module – displayed generated explanatory content.
- Quiz module – generated multiple-choice questions and displayed a score.
- Summary module – produced a topic summary.
- Learning Path module – generated a structured path for Postgres.
- Frontend navigation – demonstrated movement between available modules.

9.3 Sample Learning Path Test

The Postgres test used Intermediate as the current level and “Understand the topic well” as the learning goal. The generated path included advanced querying and joins, indexing and query performance, and data integrity and constraints.

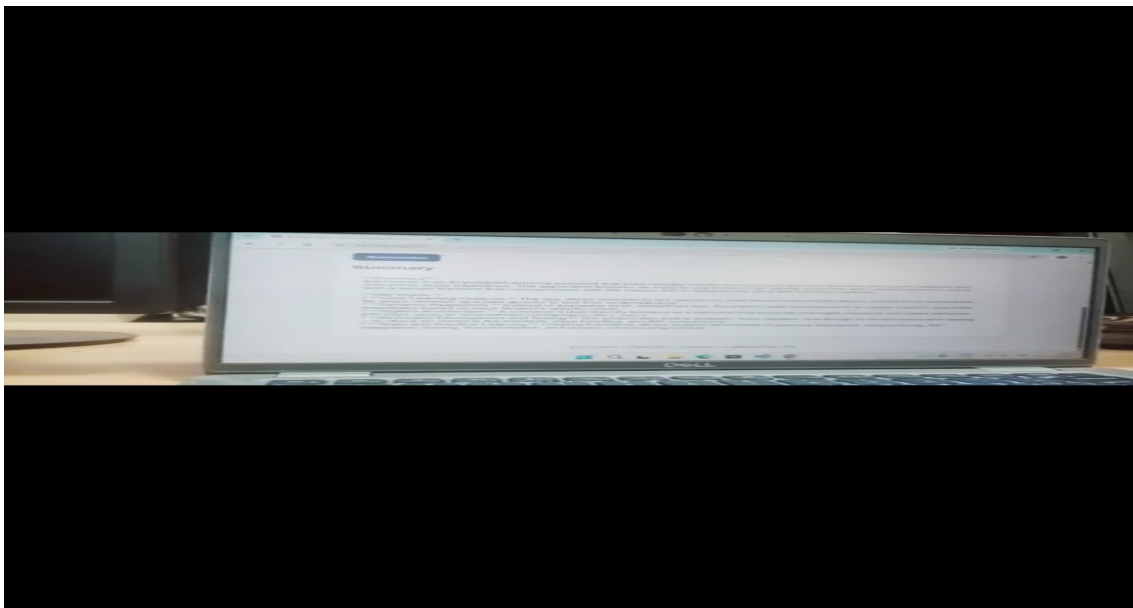


Figure 9: Summary/learning output displayed by the application

10. EPIC CONCLUSION

The Google Gemini Powered Learning Assistant demonstrates an end-to-end AI-assisted learning workflow. The project combines a web interface, FastAPI backend services and Google Gemini-based generation to provide interactive learning support.

The demonstrated modules cover question answering, explanation, quiz practice, summarization and learning-path generation. The supplied screenshots show the system producing outputs and displaying them through the web interface.

11. FUTURE ENHANCEMENT

- Add more learning modules and subject-specific tools.
- Provide user accounts and learning history.
- Store quiz scores and track progress over time.
- Add difficulty levels and adaptive question generation.
- Improve personalization of learning paths.
- Add richer visual explanations and interactive examples.
- Improve validation, error handling and API security.
- Support broader deployment and automated testing.

12. TEAM DETAILS

Role	Name
Team Lead	Santhiya. N
Team Member 1	Dharshini. R
Team Member 2	Nashrin. S
Team Member 3	Santhana Sharvani. S
Team Member 4	Swathi Lakshmi. S

This documentation is prepared using the supplied project reference documentation, team details, and the nine output screenshots provided for the Learning Assistant project.